

Radiation as a photon gas

Consider a box of volume V whose walls are maintained at a fixed temperature T . In such a situation, the walls continuously absorb and emit photons (quanta of electromagnetic radiation). These photons are bosons and since their number is not conserved here, they are distributed according to the Bose-Einstein distribution formula with zero chemical potential, i.e., the Planck distribution.

The number of photons at energy E is given by,

$$n_E = \frac{1}{e^{\beta E} - 1}$$

Since photons come in two independent polarizations (x and y or clockwise & anticlockwise),

$$n_E = \frac{2}{e^{\beta E} - 1}$$

In $\vec{k}$ -space this becomes,

$$n_{\vec{k}} = \frac{2}{e^{\beta E_{\vec{k}}} - 1}$$

Since the photons are confined to the volume V , the $\vec{k}$ values are discretized and for a cubic box, each component of $\vec{k}$ increases in steps of $2\pi/L$ (where $V=L^3$)

As seen earlier,

$$\sum_{\vec{k}} \rightarrow \frac{V}{(2\pi)^3} \int d^3 k = \frac{V}{(2\pi)^3} \int 4\pi k^2 dk \quad (\text{spherical symmetry})$$

∴ The number density of photons as a function of k per unit volume is,

$$\begin{aligned} P_k &= \frac{4\pi k^2}{(2\pi)^3} \frac{2}{e^{\beta E_k} - 1} \\ &= \frac{k^2}{\pi^2 (e^{\beta E_k} - 1)} \end{aligned}$$

$P_k dk$ = Number of photons per unit volume with wavevector in the range $[k, k+dk]$

We can express this in terms of other variables.

$$\textcircled{1} \quad E = \hbar\omega = \hbar ck$$

$$\frac{\omega}{k} = c$$

Writing $P_E dE = P_k dk,$

$$P_E = \frac{P_k}{dE/dk} = \frac{k^2}{\hbar c \pi^2 (e^{\beta E} - 1)}$$

$$\Rightarrow P_E = \frac{1}{\pi^2 (\hbar c)^3} \frac{E^2}{e^{\beta E} - 1}$$

The energy density of photons written as a function of E is,

$$J_E = E P_E = \frac{1}{\pi^2 (\hbar c)^3} \frac{E^3}{e^{\beta E} - 1}$$

$J_E dE$ = energy of photons per unit volume with energies in $[E, E+dE]$

$$\textcircled{2} \quad P_\omega d\omega = P_E dE$$

$$\Rightarrow P_\omega = \hbar P_E$$

$$= \frac{\hbar}{\pi^2 (\hbar c)^3} \frac{(\hbar\omega)^2}{e^{\beta E} - 1}$$

$$= \frac{1}{\pi^2 c^3} \frac{\omega^2}{e^{\beta \hbar \omega} - 1}$$

$\Rightarrow$ Energy density of photons as a function of ω is,

$$J_\omega = \hbar \omega P_\omega$$

$$= \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta \hbar \omega} - 1}$$

Planck's law for black body radiation.

$$\textcircled{3} \quad \text{Let } \eta \equiv \beta \hbar \omega$$

$$\begin{aligned} \Rightarrow J_\eta &= \frac{J_\omega}{d\eta/d\omega} = \frac{J_\omega}{\beta \hbar} = \frac{1}{\beta \hbar} \frac{\hbar}{\pi^2 c^3} \frac{(\eta/\beta \hbar)^3}{e^{\beta \hbar \omega} - 1} \\ &= \frac{\hbar}{\pi^2 c^3} \left(\frac{k_B T}{\hbar} \right)^4 \frac{\eta^3}{e^\eta - 1} \end{aligned}$$

The peak in energy density occurs at the value of η which maximizes $\frac{\eta^3}{e^\eta - 1}$. If this value is η_0 , then,

For two different temperatures, the maximum energy densities are obtained at angular frequencies,

$$\frac{\hbar\omega_1}{k_B T_1} = \frac{\hbar\omega_2}{k_B T_2} = \eta_0 \approx 2.82$$

$$\Rightarrow \omega^{\max}(T) = \frac{\eta_0 k_B}{\hbar} T \quad (\text{Related to Wien's displacement law})$$

The energy per unit volume due to all photons is

$$\begin{aligned} u(T) &= \frac{\hbar}{\pi^2 c^3} \left(\frac{k_B T}{\hbar} \right)^4 \int_0^\infty \frac{\eta^3}{e^\eta - 1} d\eta \\ &= \frac{\hbar}{\pi^2 c^3} \left(\frac{k_B T}{\hbar} \right)^4 \frac{\pi^4}{15} \\ &= \frac{\pi^2 k_B^4}{15 \hbar^3 c^3} T^4 \quad (\text{Stefan-Boltzmann Law}) \end{aligned}$$

$$(4) \quad k = \frac{2\pi}{\lambda}, \quad E = \frac{hc}{\lambda}$$

$$\Rightarrow P_\lambda = P_k \left| \frac{dk}{d\lambda} \right| = \frac{2\pi}{\lambda^2} \frac{k^2}{\pi^2 (e^{\beta E_k} - 1)} = \frac{8\pi}{\lambda^4 (e^{\beta hc/\lambda} - 1)}$$

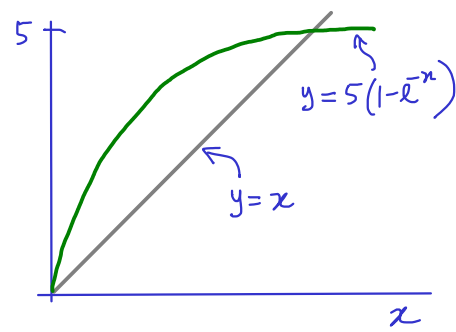
$$\begin{aligned} \Rightarrow J_\lambda &= \frac{hc}{\lambda} P_\lambda \\ &= \frac{8\pi hc}{\lambda^5 (e^{\beta hc/\lambda} - 1)} \quad \text{-- Planck's law as a function of wavelength} \end{aligned}$$

J_λ is maximum when $\lambda^5 (e^{\beta hc/\lambda} - 1)$ is minimum.

$$\Rightarrow 5\lambda_m^4 (e^{\beta hc/\lambda_m} - 1) - \lambda_m^5 \frac{\beta hc}{\lambda_m^2} e^{\beta hc/\lambda_m} = 0$$

$$\Rightarrow \frac{\beta hc}{\lambda_m} = 5 (1 - e^{-\beta hc/\lambda_m})$$

Since e^{-5} is a small number,
 if $\frac{\beta hc}{\lambda_m} = 5$, then both LHS and RHS are
 almost equal to 5.



$$\therefore \frac{\beta hc}{\lambda_m} = 5 + \epsilon \quad \text{where } \epsilon \ll 5$$

$$\Rightarrow 5 + \epsilon = 5(1 - e^{-(5+\epsilon)})$$

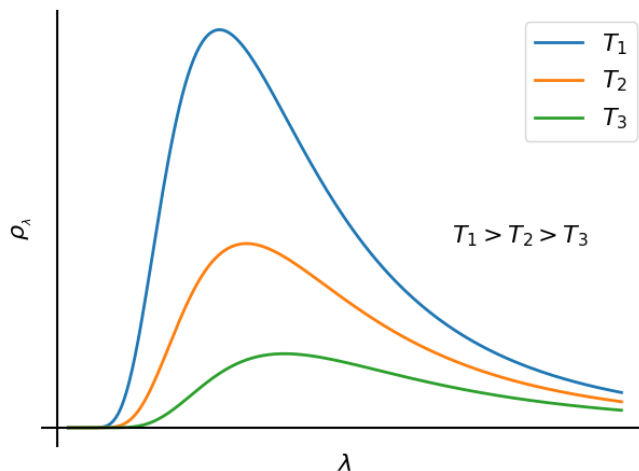
$$\Rightarrow \epsilon = -5e^{-5}e^{-\epsilon} \approx -5e^{-5}(1 - \epsilon)$$

$$\Rightarrow \epsilon \approx -\frac{5e^{-5}}{1 - 5e^{-5}} = -\frac{5}{e^5 - 5}$$

$$\Rightarrow \frac{\beta hc}{\lambda_m} \approx 5 \left[1 - \frac{1}{e^5 - 5} \right] = 5 \left(\frac{e^5 - 6}{e^5 - 5} \right)$$

$$\Rightarrow \lambda_m T \approx \frac{1}{5} \left(\frac{e^5 - 5}{e^5 - 6} \right) \frac{hc}{k_B} = 2.8978 \times 10^{-3} \text{ K}\cdot\text{m}$$

↪ Wien's displacement law



$$\textcircled{5} \quad E = h\nu \Rightarrow \mathcal{J}_\nu = h \mathcal{J}_E = \frac{1}{\pi^2 (hc)^3} \frac{h^4 \nu^3}{e^{\beta h \nu} - 1} = \frac{8\pi h}{c^3} \frac{\nu^3}{e^{\beta h \nu} - 1}$$

Radiation Pressure

The internal energy of an ideal gas (Classical, Bose, or, Fermi) is related to the pressure by the following relation.

$$U = \frac{3}{2} PV \Rightarrow u = \frac{3}{2} p$$

For radiation, this is not valid since $\epsilon_{\vec{k}} = \hbar c k \neq \frac{\hbar^2 k^2}{2m}$.

$$U = \sum_{\vec{k}} n_{\vec{k}} \epsilon_{\vec{k}} \quad , \quad \text{and,} \quad P = \sum_{\vec{k}} n_{\vec{k}} \left(-\frac{\partial \epsilon_{\vec{k}}}{\partial V} \right)$$

$$\begin{aligned} \epsilon_{\vec{k}} &= \hbar c \sqrt{k_x^2 + k_y^2 + k_z^2} \\ &= \frac{2\pi\hbar c}{L} \sqrt{n_x^2 + n_y^2 + n_z^2} \\ &= \frac{\hbar c}{V^{1/3}} \sqrt{n_x^2 + n_y^2 + n_z^2} \end{aligned}$$

[For radiation in a cubic box of edge L]

$$\Rightarrow -\frac{\partial \epsilon_{\vec{k}}}{\partial V} = \frac{\hbar c}{3V^{4/3}} \sqrt{n_x^2 + n_y^2 + n_z^2} = \frac{\epsilon_{\vec{k}}}{3V}$$

$$\Rightarrow P = \frac{U}{3V} \quad \Rightarrow \quad U = 3PV \quad \text{or} \quad \boxed{u = 3P} \quad \Rightarrow \quad P = \frac{u}{3}$$